

Daksh Patel

AI/ML Engineer

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Summary

Experienced AI/ML Engineer with 3+ years of strong background in designing, developing, and deploying machine learning models and data pipelines. Proficient in Python, deep learning frameworks, cloud platforms, and scalable model deployment. Skilled in data preprocessing, feature engineering, and model optimization to drive accurate and efficient AI solutions. Adept at collaborating with cross-functional teams to deliver impactful technology solutions aligned with business goals and operational needs.

Technical Skills

- **Programming & Frameworks:** Python, TensorFlow, Keras, PyTorch, PyTorch Geometric, OpenCV, Tesseract, Streamlit, Flask
- **Machine Learning & AI:** Reinforcement Learning, Graph Neural Networks (GAT), CNN, RNN, OCR, Feature Engineering, Hyperparameter Tuning, Bayesian Optimization, PCA, Stratified K-Fold Cross-Validation, Automated ML Pipelines, Ray Tune
- **Data Engineering & Processing:** Pandas, Apache Spark, Azure Databricks, Azure Event Hubs, Data Cleaning & Normalization
- **Cloud & Deployment:** AWS S3, AWS SageMaker, Azure Machine Learning, Azure Kubernetes Service (AKS), Kubernetes, Docker
- **Monitoring & Tools:** Prometheus, Grafana, RESTful APIs
- **Collaboration:** Agile Requirement Gathering, Cross-Functional Teamwork with Product, Risk, DevOps, MLOps, Cybersecurity

Professional Experience

AI/ML Engineer, M&T Bank

03/2024 – Present | Remote, USA

- Worked on AI-Driven Wealth Management Advisor, collaborating with product managers, financial analysts, and data engineers. Conducted requirement gathering sessions to build personalized portfolio recommendations based on user profiles, market trends, and risk tolerance.
- Collaborated with data engineering to collect, preprocess, and store investment and market data on AWS S3. Used Python and Pandas for cleaning, normalization, and feature extraction, preparing a high-quality dataset with 98% data completeness for model training.
- Developed reinforcement learning models in TensorFlow and OpenAI Gym, simulating portfolio rebalancing. Designed state-action spaces and reward functions optimizing for maximum returns and risk-adjusted metrics such as Sharpe Ratio and F1-score of 0.87.
- Applied feature engineering including client segmentation via clustering, risk factor encoding, and temporal market indicators. Used AWS SageMaker for scalable training and hyperparameter tuning, improving AUC by 12% and overall portfolio prediction accuracy.
- Tuned hyperparameters and reward thresholds via Bayesian optimization. Conducted extensive A/B testing across client cohorts, collaborating with UX and product teams to measure uplift in adoption and client satisfaction by 25%.
- Validated model performance and compliance with risk teams. Partnered with DevOps and MLOps teams to containerize the model using Docker and deploy it on Kubernetes. Exposed RESTful APIs via Flask for integration with digital banking platforms.

AI/ML Engineer, Barclays

08/2021 – 12/2022 | Remote, India

- Settled a real-time fraud detection system using Graph Neural Networks in alliance with data science, fraud risk, and compliance teams. Led requirement-gathering sessions and modeled transactional graphs to identify 75% more complex fraud scenarios.
- Designed and implemented a scalable data pipeline using Azure Event Hubs for streaming ingestion and Apache Spark on Azure Databricks for real-time data processing. Cleaned 60+ million daily transactions to reach 99.8% usable training data.
- Built custom Graph Attention Network (GAT) models using PyTorch Geometric and NetworkX. Trained models on Azure Machine Learning service with automated ML pipelines, improving detection accuracy by 38% on labeled transactional graph data.
- Applied feature engineering methods such as edge-based scoring, transaction frequency profiling, and time-window aggregation. Reduced feature space by 30% using PCA while preserving 92% information, in sync with the analytics and modeling teams.
- Conducted model validation using stratified k-fold cross-validation and live A/B testing. Tuned models with Ray Tune to boost precision by 42% and cut false positives by 22% in real-time fraud detection scenarios.
- Collaborated with DevOps, MLOps, cybersecurity, and platform engineering teams for deployment. Containerized using Docker, deployed via Kubernetes on Azure Kubernetes Service, and implemented real-time monitoring using Prometheus and Grafana.

Machine Learning Engineer, Barclays

01/2021 – 08/2021 | Remote, India

- Collaborated with compliance, onboarding, and backend teams to gather requirements and define KYC document workflows. Labeled datasets across PAN, Aadhaar, and Passport formats for model training, testing, and secure file classification automation.
- Built OCR pipeline using Tesseract and OpenCV for preprocessing scanned documents including noise removal, edge detection, and skew correction. Extracted text blocks and metadata, transforming raw images into structured input for classification models.
- Designed and trained CNN and RNN hybrid models in TensorFlow/Keras for multi-label classification of document types. Tuned hyperparameters and validated using stratified k-fold cross-validation, achieving 91% macro F1-score across varied file resolutions.
- Deployed model using Streamlit for demo UI and integrated production-ready API into internal KYC verification system. Collaborated with DevOps and MLOps teams to containerize with Docker, reducing verification time by 70% and onboarding friction.

Education

Arizona State University

Master of Science in Information Technology - Ira A. Fulton Schools of Engineering

LDRP Institute of Technology and Research

Bachelor of Technology in Computer Engineering

Tempe, AZ, USA

01/2023 – 12/2024

Gandhinagar, India

08/2018 – 05/2022